

EN25QW16APR1L1
16 Megabit wide voltage Serial Flash Memory with 4Kbyte Uniform Sector
Physical Specification

Parameter	Value	Unit
Wafer diameter	12	inch
Die Thickness (Without grinding)	780 ± 25	μm
Die Thickness (Minimum grinding thickness)	80	μm
Die size (Without scribe line)	1235 x 1205	μm
Gross die / wafer	40561	ea
Scribe Line Width	70	um
Pad size	58 x 58	um
Pad Material and Thickness	AlCu = 14500A	-
Die Backside	No metal, May be grounded	-
Passivation	TEOS Oxide/Nitride	-

Ordering Information

Product ID	DC	Max Freq.	V _{CC}	Operation junction temperature (T _J)
EN25QW16APR1L1	1	104Mhz 80Mhz 33Mhz 60MHz (Suspend/Resume is not supported)	2.3~3.6V 1.8~2.3V 1.65~1.8V 1.65~1.8V	-40°C ~ +85°C
	0	66Mhz 33Mhz 60MHz (Suspend/Resume is not supported)	1.8~3.6V 1.65~1.8V 1.65~1.8V	

Key Functional & Electrical Specification

For the detailed product specification (including product features, functions, AC and DC performance), please refer to the data sheet of EN25QW16A (2P). If there is any conflict between KGD and discrete part datasheets, the specification in this KGD datasheet is recommended.

Processing

Do not expose Die products to ultraviolet light or process them at temperatures greater than 250°C. Failure to adhere to these handling instructions will result in irreparable damage to the devices. For best yield, ESMT recommends assembly in a Class 10K clean room with 30% to 60% relative humidity.

Storage

Store at a maximum temperature of 30°C in a nitrogen purged cabinet or vacuum-sealed bag. Observe all standard ESD handling procedures.

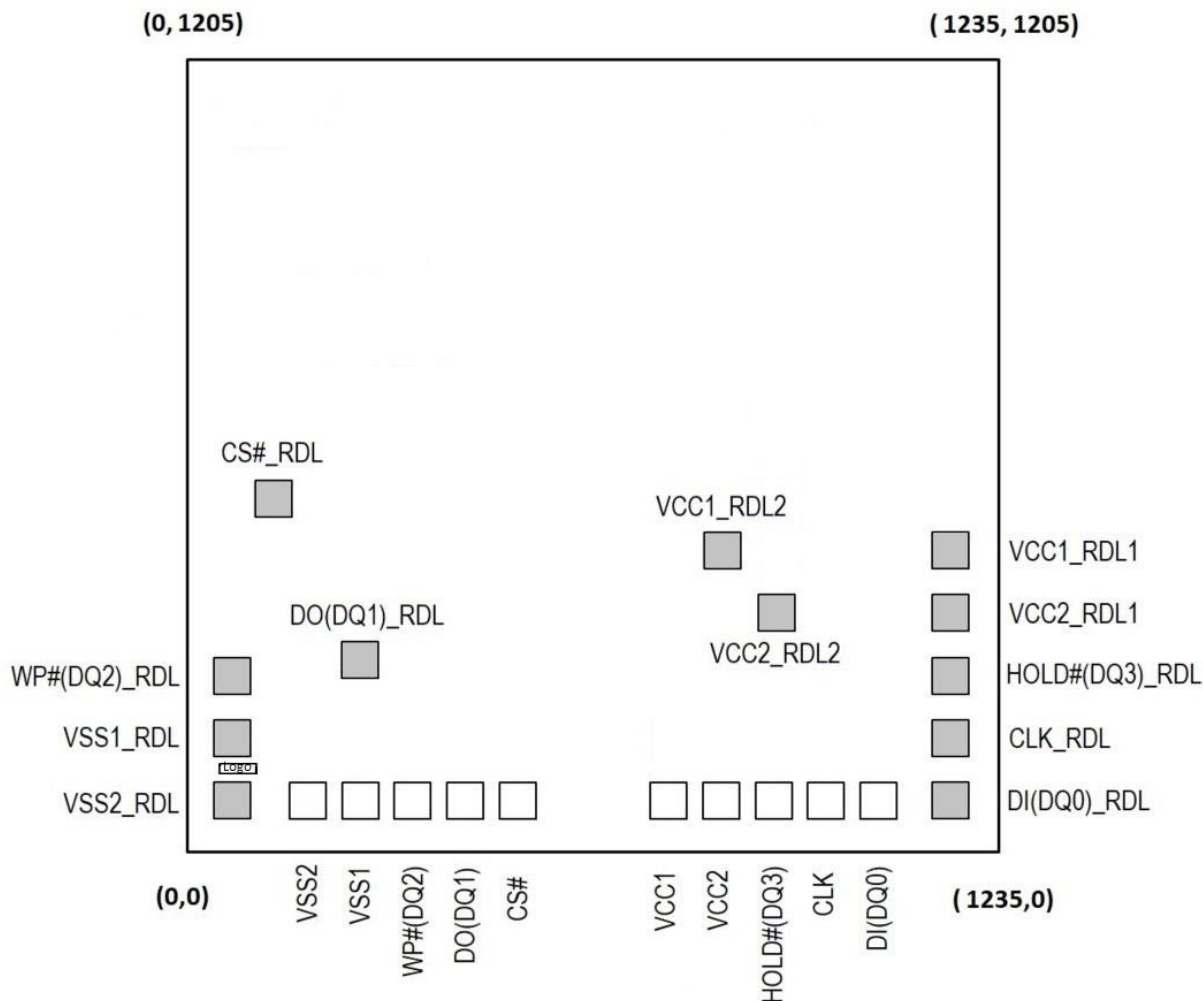
Conditions of Die use

ESMT guarantee unpackaged Die against defective materials or workmanship for a period of one (1) year from the date of shipment. This warranty extends only to the first purchaser of the Die. Buyer assumes full responsibility to ensure compliance with the appropriate handling ,assembly and processing of unpackaged Die (including but not limited to proper Die preparation ,Die attach ,wire bonding and related assembly and test activities), and compliance with all guidelines set forth in ESMT's specifications for Die product.

ESMT assumes no responsibility for environmental effects on Die or for any activity of Buyer or a third party that damages the Die due to improper use, abuse, negligence, improper installation, accident, loss, damage in transit, or unauthorized repair or alteration by a person or entity other than ESMT.

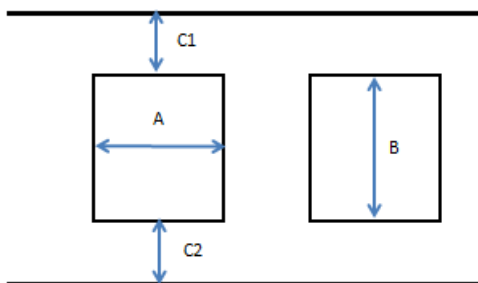
Die are not designed or authorized for use as components in life support appliances, devices or systems where malfunction of the Die can reasonably be expected to result in a personal injury. Buyer's use of Die product for use in life support applications is at Buyer's own risk and Buyer agrees to fully indemnify ESMT for any damages resulting in such use or sale.

Die Pad Locations

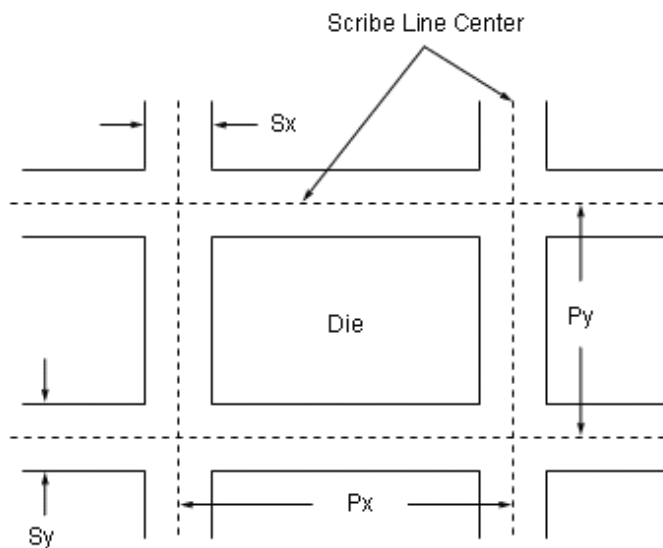


TEST KEY INFORMATION (in scribe line)

Parameter	Value	Unit
Test key pad size (A x B)	70.23 x 60.2	μm
Test key pad metal thickness	1.45	μm
Test key pad to seal ring distance (C1 / C2)	4.9 / 4.9	μm
Test key distribution	Center	-

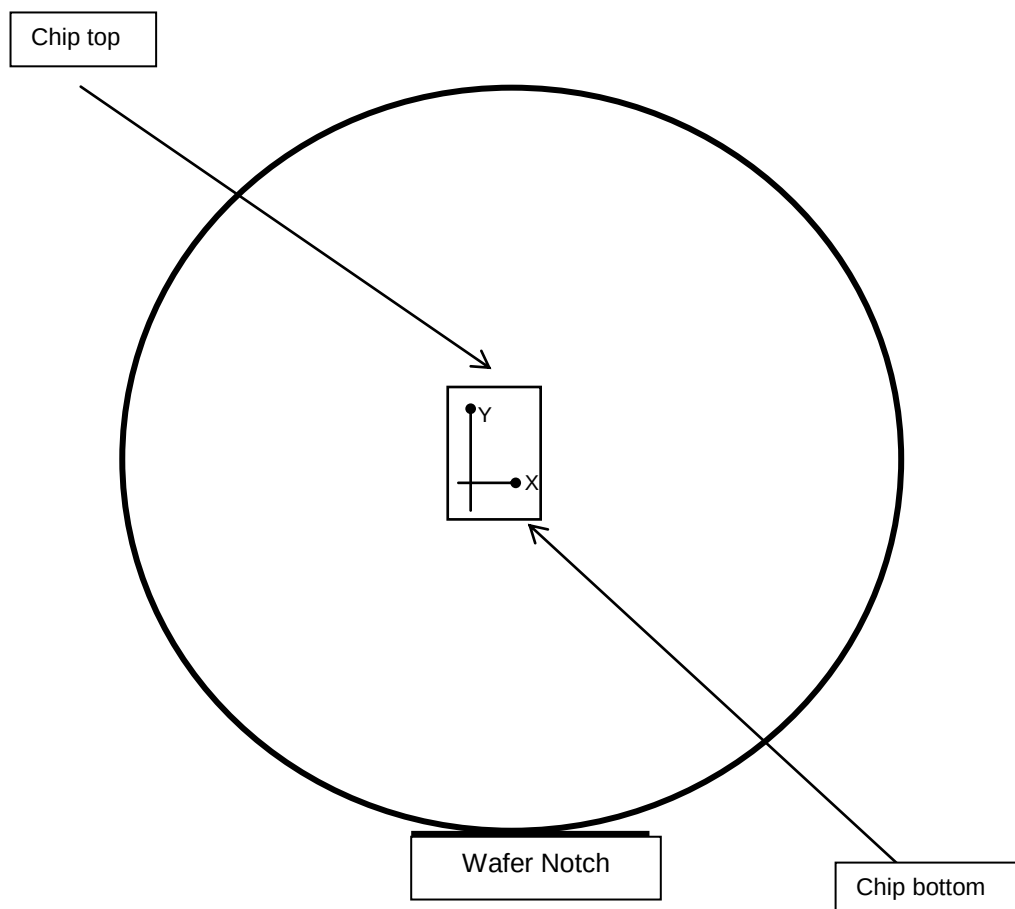


Scribe Line Illustration



Pitch die size: P_x : 1305 μm , P_y : 1275 μm
 Scribe line width: $S_x = S_y = 70 \mu\text{m}$

Wafer Chip Locations



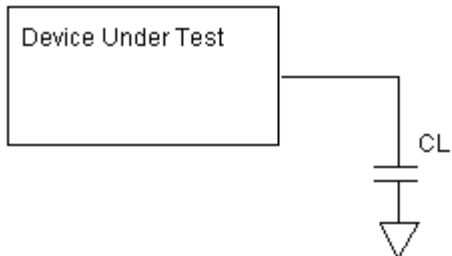
Pad Coordinates

ORIGINAL PAD					
Pad Name	X	Y	Pad Name	X	Y
CS#	503.35	78.29	VCC1	732.43	78.29
DO(DQ1)	423.35	78.29	VCC2	812.43	78.29
WP#(DQ2)	343.35	78.29	HOLD#(DQ3)	892.43	78.29
VSS1	263.35	78.29	CLK	972.43	78.29
VSS2	183.35	78.29	DI(DQ0)	1052.43	78.29
-	-	-	LOGO	77	136.5

RDL PAD					
Pad Name	X	Y	Pad Name	X	Y
CS#_RDL	131.5	538.4	VCC1_RDL1	1161.5	458.29
DO(DQ1)_RDL	263.345	292.09	VCC1_RDL2	814.5	458.29
WP#(DQ2)_RDL	68.5	268.29	VCC2_RDL1	1161.5	363.29
VSS1_RDL	68.5	173.29	VCC2_RDL2	896.735	363.29
VSS2_RDL	68.5	78.29	HOLD#(DQ3)_RDL	1161.5	268.29
CLK_RDL	1161.5	173.29	DI(DQ0)_RDL	1161.5	78.29

Pad Group.	Count	Rule
VSS1/ VSS1_RDL	2	Optional a pad connect to GND
VSS2/ VSS2_RDL	2	Both pads must connect to GND
VCC1/ VCC1_RDL1/ VCC1_RDL2	3	Optional a pad connect to VCC
VCC2/ VCC2_RDL1/ VCC2_RDL2	3	Optional a pad connect to VCC

Test Conditions



Test Specifications

Test Conditions		Unit
Output Load Capacitance, C_L	30	pF
Input Rise and Fall times	5	ns
Input Pulse Levels	$0.2V_{CC}$ to $0.8V_{CC}$	V
Input timing measurement reference levels	$0.3V_{CC}$ to $0.7V_{CC}$	V
Output timing measurement reference levels	$V_{CC} / 2$	V

Revisions List

Revision No	Description	Date
1.0	Initial Release	2024.07.26

Important Notice

All rights reserved.

No part of this document may be reproduced or duplicated in any form or by any means without the prior permission of ESMT.

The contents contained in this document are believed to be accurate at the time of publication. ESMT assumes no responsibility for any error in this document, and reserves the right to change the products or specification in this document without notice.

The information contained herein is presented only as a guide or examples for the application of our products. No responsibility is assumed by ESMT for any infringement of patents, copyrights, or other intellectual property rights of third parties which may result from its use. No license, either express, implied or otherwise, is granted under any patents, copyrights or other intellectual property rights of ESMT or others.

Any semiconductor devices may have inherently a certain rate of failure. To minimize risks associated with customer's application, adequate design and operating safeguards against injury, damage, or loss from such failure, should be provided by the customer when making application designs.

ESMT's products are not authorized for use in critical applications such as, but not limited to, life support devices or system, where failure or abnormal operation may directly affect human lives or cause physical injury or property damage. If products described here are to be used for such kinds of application, purchaser must do its own quality assurance testing appropriate to such applications.